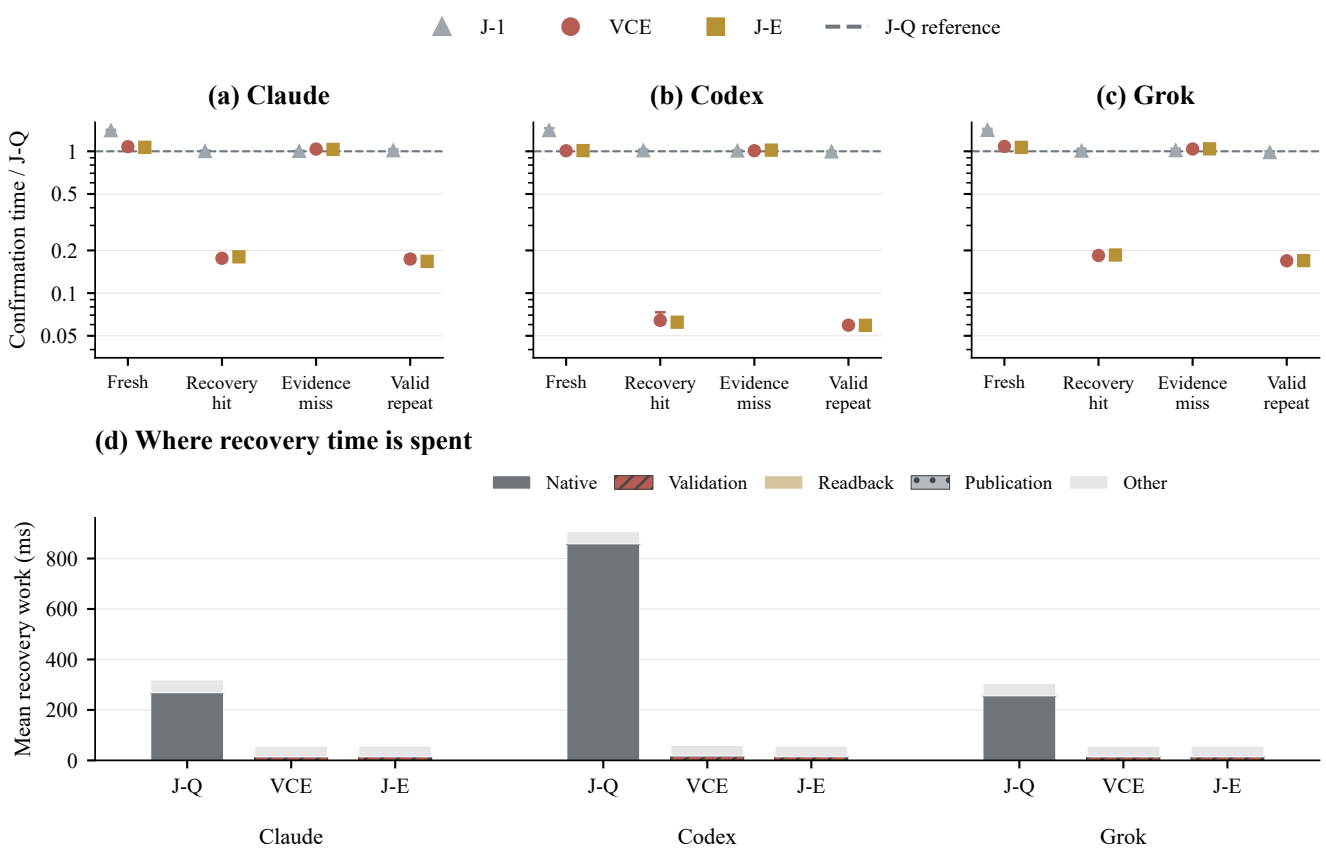

J-1
VCE
J-E
J-Q reference
(a) Claude
(b) Codex
(c) Grok
Confirmation time / J-Q
1
0.5
0.2
0.1
0.05
Fresh
Recovery hit
Evidence miss
Valid repeat
(d) Where recovery time is spent
Native
Validation
Readback
Publication
Other
Mean recovery work (ms)
800
600
400
200
0
J-Q
VCE
J-E
Claude
Codex
Grok